Programming Fundamentals

LAB-05 Submission

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LAB Task-01:

Question: Write a C++ program to print Table of any number.

Task 1: Write a C++ Program to Print Table of any number.

Code:

```
#include <iostream>
using namespace std;
int main() {
.  int num;
.  cout << "Enter any number: ";
.  cin >> num;
.  for (int i = 1; i <= 10; i++)
.  cout << num << " * " << i << " = " << num * i << endl;
}
}  return 0;
```

Output:

```
Enter any number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

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LAB Task-02:

Question: Write a C++ program to find Power

Task 2: Power of any number usIng for loop

Code:

```
#include <iostream>
using namespace std;
int main() {
    int base, Power, result = 1;
    cout << "Enter base: ";
    cin >> base;
    cout << "Enter Power: ";
    cin >> Power;
    for (int i = 1; i <= Power; i++) {
        result = result * base;
    }
    cout << "result = " << result;
    return 0;
}
```

Output:

```
Enter base: 2
Enter Power: 3
result = 8
```

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Task 3: Sum of Natural numbers Using Loop

Code (Corrected):

```
#include <iostream>
using namespace std;
int main() {
    int n, Sum = 0;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        Sum = Sum + i;
    }
    cout << "Sum = " << Sum;
    return 0;
}
```

Output:

```
Enter a number: 5
Sum = 15
[Process completed]
```

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LAB Task-04:

Question: C++ Program to find Sum of digits of a Number

Task 4: Sum of digits of a Number

Code:

```
#include <iostream>
using namespace std;
int main() {
    int num, digit, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    while (num > 0) {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }
    cout << "Sum of digits = " << sum;
    return 0;
```

Output:

```
Enter a number: 350
Sum of digits = 8
```

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LAB Task-05:

Question: C++ Program to find Factorial of a number

Task 5: Factorial of a number

Code:

```
#include <iostream>
using namespace std;
int main() {
    int num, fact = 1;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= num; i++) {
        fact = fact * i;
    }
    cout << "Factorial = " << fact;
    return 0;
}
```

Output:

```
Enter a number: 4
Factorial = 24
```

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LAB Task-06:

Question: C++ Program to Print the Pattern

Pattern:

```
*
**
***
****
*****
******
*******
```

Task 6: Print the Star Pattern

Code:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 7; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
```

Output:

```
*
**
***
****
*****
******
*******
```

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LAB Task-07:

Question: Print Total Days in a Month Using Switch Case

Task 7: Days In a Month

Code:

```
#include <iostream>
using namespace std;
int main() {
    int month;
    cout << "Enter month number (1-12): ";
    cin >> month;
    switch(month) {
        case 1: case 3: case 5:
        case 7: case 8: case 10:
        case 12:
            cout << "31 days";
            break;
        case 4: case 6: case 9: case 11:
            cout << "30 days";
            break;
        case 2:
            cout << "28 or 29 days";
            break;
        default:
            cout << "Invalid month number";
```

Output:

```
Enter month number (1-12): 4
30 days
```

Explanation:

- Month 1,3,5,7,8,10,12 => 31 days
- Month 4,6,9,11 => 30 days
- Month 2 => 28 or 29 days
- Invalid => Invalid month number

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END

Thank You!

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